Sahaj Bhadja

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Summary

Software engineer with 2+ years building AI-enabled services, secure APIs, distributed workflows, and cloud automation using Python, Java, C++, LLM APIs, AWS, Azure, Docker, Kubernetes, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant Azure SQL and Blob Storage services with row-level isolation for 500+ users, optimizing indexed access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI orchestration for 5,000+ monthly AI jobs with explicit states, validation, reconciliation, monitoring, and failure isolation, achieving 99%+ reliability across recurring production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Integrated GPT-4o, Claude, Qwen, and OCR into Flask services handling 10,000+ daily grading requests, securely validating model outputs and reducing instructor feedback turnaround by 60%.
- Designed concurrent processing for 1,000+ uploaded files using secure ingestion, multithreading, AWS RDS, and fault-tolerant execution, isolating failures while maintaining consistently stable API performance during peaks.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Automated nightly Jenkins pipelines for builds, VM installation, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, producing nightly coverage summaries and actionable failure logs.
- Optimized REST API and ETL workflows using object-oriented patterns, AWS Glue, S3, and SQL, improving response times by 30%, pipeline efficiency by 68%, and cross-platform systems maintainability.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a modular MERN application with REST APIs, WebSockets, and eight services supporting 2M+ synchronized records while translating defined requirements into responsive customer and inventory workflows reliably.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving integration consistency, defect resolution, and production delivery reliability for engineering teams.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for a booking application supporting scheduling, payments, and status updates across 50,000+ records with synchronized Firebase-backed workflows, validation controls, and clear operational state tracking.
- Implemented webhook-based payment synchronization and root-cause troubleshooting, reducing manual handling by 70% while improving transaction visibility, error recovery, and responsiveness across distributed application components during production operations.

Skills

Programming & APIs: Python, Java, C++, C, JavaScript, TypeScript, SQL, Bash, FastAPI, Flask, Node.js, REST APIs, WebSockets, microservices
AI & Security: LLM APIs, GPT-4o, Claude, Qwen, prompt design, output evaluation, embeddings, vector search, authentication, authorization, row-level security, secure file ingestion
Cloud & DevOps: AWS, Azure, GCP, Amazon S3, AWS Glue, AWS RDS, Azure SQL, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, monitoring
Engineering: PyTest, Mockito, unit testing, integration testing, code reviews, Agile/Scrum, distributed systems, failure isolation, debugging, operational reliability, performance optimization, networking fundamentals

Projects

Big-Brain Live | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built a secure AI document platform for 200+ users with authenticated ingestion, vector indexing, semantic retrieval, and collaborative workflows, maintaining 95% query accuracy across 50,000+ searchable documents.

Blockchain Authentication Paper | *Security, IoT, Access Control*
- Evaluated secure IoT device authentication and role-based access across four permission tiers, designing decentralized verification flows that reduced repeated key-verification steps by 50% across modeled security scenarios.

Linguistix | *Python, AWS ECS/Fargate, Docker, CodePipeline, CI/CD*
- Developed a containerized data-processing API on AWS with automated CodePipeline releases, executable validation checks, and Docker builds, achieving 87% accuracy and production-grade reliability across repeatable integration environments.